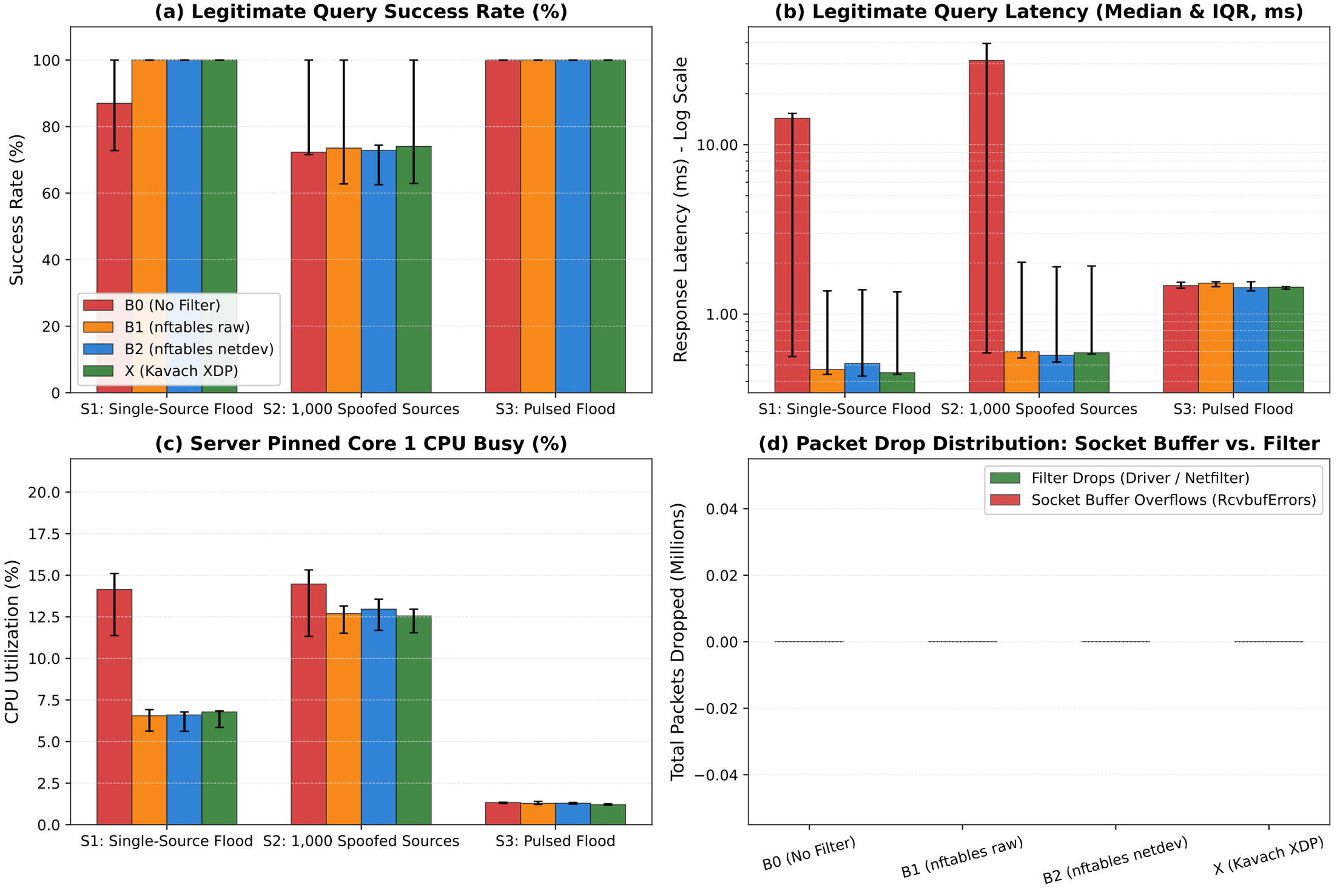
Kavach Physical Lab Benchmark: 108 Runs Matrix (HP ProDesk 4 / Intel e1000e)
(a) Legitimate Query Success Rate (%)
Success Rate (%)
0
20
40
60
80
100
B0 (No Filter)
B1 (nftables raw)
B2 (nftables netdev)
X (Kavach XDP)
S1: Single-Source Flood
S2: 1,000 Spoofed Sources
S3: Pulsed Flood
(b) Legitimate Query Latency (Median & IQR, ms)
Response Latency (ms) - Log Scale
10.00
1.00
S1: Single-Source Flood
S2: 1,000 Spoofed Sources
S3: Pulsed Flood
(c) Server Pinned Core 1 CPU Busy (%)
CPU Utilization (%)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
S1: Single-Source Flood
S2: 1,000 Spoofed Sources
S3: Pulsed Flood
(d) Packet Drop Distribution: Socket Buffer vs. Filter
Filter Drops (Driver / Netfilter)
Socket Buffer Overflows (RcvbufErrors)
Total Packets Dropped (Millions)
0.04
0.02
0.00
−0.02
−0.04
B0 (No Filter)
B1 (nftables raw)
B2 (nftables netdev)
X (Kavach XDP)